

Customer Behaviour Analysis Report

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Abstract

This report presents an end-to-end customer behaviour analysis using structured data analytics techniques. The study focuses on understanding customer shopping patterns and preferences through data extraction using SQL, exploratory data analysis using Python, and visualization through Power BI dashboards.

1. Introduction

Customer behaviour analysis plays a critical role in data-driven decision-making. By analyzing purchasing behaviour, organizations can improve customer engagement, optimize product strategies, and enhance overall business performance. This project demonstrates a complete analytics pipeline applied to customer shopping data.

2. Objectives

- To analyze customer shopping behaviour patterns
- To perform exploratory data analysis using Python
- To visualize insights using Power BI dashboards

3. Data Description

The dataset consists of customer shopping behaviour records, including purchase-related attributes. The data was stored in a PostgreSQL database and retrieved using structured SQL queries for further analysis.

4. Methodology

The methodology followed an end-to-end analytics workflow. Data was first extracted from PostgreSQL using SQL queries. The extracted data was then analyzed using Python libraries such as pandas and numpy. Exploratory Data Analysis (EDA) was performed to identify trends and patterns. Finally, insights were visualized using Power BI.

5. Tools and Technologies

PostgreSQL was used for database management and querying. Python and Jupyter Notebook were used for data analysis. Power BI was used to create interactive dashboards for visualization.

6. Results and Insights

The analysis revealed clear patterns in customer purchasing behaviour, including product preferences and trends. Visualization through Power BI enabled easy interpretation of insights, supporting data-driven decision-making.

7. Conclusion

This project successfully demonstrates an end-to-end customer behaviour analysis pipeline. The integration of SQL, Python, and Power BI showcases practical data analytics skills and highlights the importance of data-driven insights in understanding customer behaviour.

References

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